

ALX Foundations: Milestone 3 Worksheet

Instructions: Provide responses to all items in the orange boxes. The worksheet consists of Sections A-E. Work on this worksheet one section at a time throughout your week, and return to Canvas after each section for the next set of content and further instructions.

SECTION A: Problem Statement

Step 1: Your GCGO

Which Grand Challenge or Great Opportunity (GCGO) do you want to play a part in addressing? (Pick one.)

As a reminder, the GCGOs are:

- Urbanization
- Education
- Infrastructure
- Healthcare
- Climate change
- Governance
- Job creation
- Agriculture
- Natural resources
- Arts, culture, and design
- Tourism
- Empowerment of women
- Regional integration
- Wildlife conservation

GCGO: Urbanization

Step 2: Describe Your Problem

You are going to take a first pass at briefly describing your chosen problem. This can be any problem that speaks to you, as long as it is a real-life occurrence that is clearly linked to your chosen GCGO, that occurs in a certain place and for certain people (and/or animals), and that it can be clearly defined.

For example, if you chose wildlife conservation as your GCGO, you might first state your chosen problem as:

There are very few white rhinos left in Kenya and they are in danger of going extinct.

Another example, if you choose infrastructure as your GCGO:

Residents of major cities in South Africa endure prolonged periods without electricity, significantly hampering their ability to generate income.

Note that this is just your first attempt stating the problem, and you don't need to quantify the problem yet. In order to get to your official problem statement (which does need to be quantifiable) first answer the following questions. The more specific your answers, the better. You may also ask Google, Wikipedia, , ChatGPT, and/or other reliable online sources to help you. Please be sure to cite (give credit to) any sources that you use.

Describe your problem using What/Who/When/Where/Why/How....

1. **What** is the problem? What is reality like because of this problem? What will reality be like if the problem continues?

Problem Description:

The problem is the high levels of traffic congestion in Nairobi, which result in residents spending excessive amounts of time commuting to their workplaces. This situation significantly affects the productivity and efficiency of commuters in the city. If the problem continues, it will lead to further deterioration of productivity, increased stress levels, and reduced quality of life for residents. As more people continue to migrate to the city, the traffic congestion problem is expected to persist and worsen, exacerbating the negative impact on daily life and hindering the city's overall development.

2. **Who** does this problem impact, directly and indirectly? Who contributes to the problem?

This problem directly impacts the residents of Nairobi and indirectly affects businesses operating within the city. Commuters who spend significant time in traffic experience reduced productivity, increased stress levels, and a diminished quality of life. Businesses also suffer from the inefficiencies caused by traffic congestion, including delayed deliveries, increased transportation costs, and reduced customer satisfaction. The problem is partly attributed to the poor planning and infrastructure management by the county government, which has contributed to the growing traffic congestion over time.

3. **When** did this problem begin? When does it occur?

The problem of traffic congestion in Nairobi has been a longstanding issue that has worsened over time. Nairobi, the capital city of Kenya, has experienced rapid urbanization and population growth, leading to increased vehicular traffic on its road networks. The problem began to emerge as the city's population expanded and economic activities intensified. The occurrence of traffic congestion is most prevalent during peak hours, specifically in the morning from 6-9 am and in the evening from 6-8 pm, commonly known as rush hour. During these times, the influx of vehicles commuting to and from work, coupled with inadequate infrastructure and road capacity, leads to severe traffic congestion throughout the city.

4. **Where** is this problem occurring? What is the context in which it occurs?

The problem of traffic congestion is occurring in the city of Nairobi, Kenya. Nairobi is the capital and largest city in the country, serving as a major economic hub and a center for commerce, administration, and cultural activities. The city's infrastructure, including its road network, has struggled to keep up with the rapid urbanization and population growth experienced in recent years. The problem of traffic congestion is particularly prevalent in key areas of the city, such as the central business district (CBD), major highways, and busy intersections. These locations witness high volumes of vehicles from residents, commuters, and commercial activities, leading to congestion and traffic gridlock. The context in which this problem occurs includes a combination of factors such as population density, inadequate road infrastructure, ineffective traffic management systems, and a lack of alternative transportation options.

5. **Why** is this a problem? What are the pain points or gaps? Why do you personally care about this problem?

The problem of traffic congestion in Nairobi is a significant issue due to its negative impact on the daily lives of residents and the overall productivity of the city. The pain points and gaps associated with this problem include:

1. Time wastage: Traffic jams result in significant time delays, causing frustration and reducing overall productivity. Commuters spend excessive time in traffic, which could be better utilized for work, leisure, or family activities.
2. Economic impact: Traffic congestion leads to increased fuel consumption, higher transportation costs, and reduced efficiency in the movement of goods and services. This negatively affects businesses, increases operational costs, and hampers economic growth.
3. Environmental implications: Traffic congestion contributes to increased air pollution, as vehicles spend more time idling in traffic. This has adverse effects on air quality, public health, and the environment.
4. Stress and health issues: The constant exposure to traffic congestion can lead to heightened stress levels and impact the mental and physical well-being of individuals. Increased anxiety, frustration, and fatigue are common outcomes of dealing with daily traffic congestion.

As an AI engineer with expertise in computer vision, I personally care about this problem because I believe technology can play a crucial role in addressing the challenges of traffic congestion. AI-powered solutions, such as intelligent traffic management systems, predictive analytics, and smart transportation infrastructure, have the potential to optimize traffic flow, reduce congestion, and improve the overall transportation experience for individuals. By leveraging AI and computer vision technologies, we can work towards creating smarter, more efficient transportation systems that enhance mobility, reduce travel times, and promote sustainable urban development.

6. **How** would reality be different if this problem were solved? (This can be your opinion.) If the problem of traffic congestion in Nairobi were effectively solved, reality would be significantly transformed:

1. Improved productivity: Commuters would experience reduced travel times, allowing them to reach their destinations faster and allocate more time to productive activities, enhancing overall efficiency.
2. Enhanced quality of life: With reduced traffic congestion, individuals would experience less stress, frustration, and fatigue associated with long commutes, leading to improved well-being and a better work-life balance.
3. Economic growth: Efficient traffic flow would enhance business operations, lower transportation costs, and attract investments, fostering economic growth and development in the city.
4. Environmental sustainability: Reduced traffic congestion would result in decreased emissions, leading to improved air quality and a healthier environment for residents.

Overall, solving the problem of traffic congestion would create a more livable, sustainable, and vibrant city, benefiting both residents and businesses alike.

Step 3: Understand and Quantify Your Problem

Next, you will conduct some basic web research to better understand, define, and quantify your problem. You will do this through a combination of Google search, Wikipedia, credible web sources, ChatGPT or other AI research tool, and your own synthesis of information from these sources. Be sure to give credit to your sources, and paraphrase (use your own words) rather than quoting directly.

7. What is the historical context for this problem? What happened in the past that contributes to the problem now?

1. Rapid urbanization: Nairobi has experienced significant population growth and urbanization over the years, leading to increased demand for transportation

infrastructure that has not kept pace with the rising population. This has resulted in congestion on road networks.

2. Limited road infrastructure: The road network in Nairobi was not designed to accommodate the current volume of vehicles, leading to bottlenecks and traffic congestion. The inadequate expansion and maintenance of roads have contributed to the problem.

3. Lack of effective public transportation: Nairobi has faced challenges in developing and maintaining an efficient public transportation system. This has resulted in a high reliance on private vehicles, further exacerbating traffic congestion.

4. Inadequate urban planning: Insufficient urban planning and zoning have led to mixed land use and the concentration of businesses and residential areas in certain parts of the city. This has resulted in traffic convergence in specific locations during peak hours.

5. Population growth and vehicle ownership: The increasing population in Nairobi, coupled with rising vehicle ownership, has put additional strain on the road infrastructure, leading to congestion.

8. What are the possible economic (money-related) reasons why this problem exists and continues?

1. Inadequate investment in transportation infrastructure: Insufficient funding and investment in expanding and upgrading the road network, public transportation systems, and traffic management infrastructure can hinder effective traffic flow and exacerbate congestion.

2. Lack of alternative transportation options: The limited availability and accessibility of affordable and efficient public transportation options may incentivize individuals to rely on private vehicles, leading to increased congestion. Insufficient investment in alternative modes of transport, such as light rail or metro systems, can contribute to this issue.

3. Economic disparity and urban sprawl: Socioeconomic disparities and urban sprawl can contribute to longer commuting distances as people move to areas with affordable

housing or seek employment opportunities. This leads to increased travel demand and congestion, as individuals may need to commute long distances to access job centers.

4. Insufficient parking facilities: Inadequate provision of parking spaces in commercial and residential areas can lead to illegal parking on roadsides, further narrowing the road space available for traffic and causing congestion.

5. Lack of congestion pricing or road user charges: The absence of congestion pricing mechanisms or road user charges, such as tolls or congestion fees, may result in limited financial disincentives for vehicle usage during peak hours. This can contribute to sustained traffic congestion.

9. What are the possible political reasons why this problem exists and continues?

1. Inadequate urban planning and governance: Poor urban planning practices, ineffective governance, and lack of long-term transportation strategies can contribute to traffic congestion. Political decisions related to infrastructure development, zoning regulations, and transportation policies may not align with the growing needs of the city.

2. Lack of coordination among relevant government agencies: Inefficient coordination and collaboration among different government agencies responsible for transportation, urban development, and traffic management can lead to disjointed efforts and suboptimal solutions to address congestion.

3. Delayed decision-making and policy implementation: Political factors such as bureaucracy, competing priorities, and frequent changes in leadership may result in delayed decision-making and implementation of transportation projects and policies. This can hinder the timely and effective mitigation of traffic congestion.

4. Influence of vested interests: Political pressures and influence from stakeholders, including car manufacturers, construction companies, and influential individuals, may shape transportation policies and infrastructure investments in favor of private vehicle usage rather than promoting sustainable and efficient alternatives.

5. Lack of political will for unpopular measures: Implementing certain measures, such as congestion pricing or restricted car usage policies, may face resistance due to potential public backlash or perceived negative impact on certain interest groups. This can limit the willingness of political leaders to take bold actions to tackle congestion.

Addressing these political reasons requires strong political commitment, effective governance structures, coordinated decision-making, and prioritization of long-term transportation planning over short-term political gains. Engaging with relevant stakeholders, considering expert advice, and implementing evidence-based policies can help alleviate traffic congestion in Nairobi.

10. What cultural beliefs and/or social norms possibly contribute to this problem?

1. Car ownership as a status symbol: In many cultures, owning a car is often associated with social status and prestige. This belief can lead to a higher demand for private vehicles, resulting in increased congestion as more people choose to drive instead of using alternative modes of transportation.
2. Individualistic mindset: Cultures that emphasize individualism and personal convenience may prioritize private vehicle usage over public transportation. This mindset can contribute to the overreliance on cars and the resulting traffic congestion, as individuals prioritize their own comfort and convenience over collective transportation solutions.
3. Resistance to change: Cultural beliefs and social norms rooted in tradition and resistance to change can impede the adoption of new transportation practices and technologies. This can hinder the implementation of innovative solutions to alleviate congestion, such as carpooling, ride-sharing, or embracing alternative modes of transportation.
4. Lack of pedestrian-friendly infrastructure: In some cultural contexts, walking or cycling may not be perceived as favorable modes of transportation. Insufficient infrastructure for pedestrians and cyclists, combined with cultural preferences for motorized transport, can discourage people from choosing sustainable alternatives and contribute to traffic congestion.
5. Time perception and punctuality: Cultural attitudes towards time and punctuality can influence commuting behaviors. If lateness is culturally tolerated or expected, individuals may be less motivated to plan their journeys efficiently, contributing to congestion during peak hours.

Addressing cultural beliefs and social norms related to transportation requires a multifaceted approach that involves education, awareness campaigns, infrastructure improvements, and promoting sustainable modes of transportation. Encouraging a

shift in cultural norms towards shared responsibility, public transportation usage, and embracing alternative modes of travel can help alleviate traffic congestion in Nairobi.

11. Who are the people potentially responsible (directly or indirectly) for creating and/or maintaining this problem?

1. Urban planners and policymakers: The decisions made by urban planners and policymakers regarding transportation infrastructure, road design, zoning regulations, and public transportation systems play a significant role in shaping the transportation landscape. Inadequate planning, lack of long-term vision, and failure to prioritize sustainable transportation options can contribute to traffic congestion.
2. Government authorities and transportation agencies: Government authorities responsible for transportation, such as transport ministries, departments, and agencies, have a direct impact on the transportation system. Inefficient management, insufficient funding, and delayed implementation of transportation projects can contribute to congestion and hinder effective traffic flow.
3. Road users: Individual drivers and commuters also play a role in creating and maintaining traffic congestion. Inadequate adherence to traffic rules and regulations, aggressive driving behavior, lack of consideration for other road users, and overreliance on private vehicles contribute to congestion.
4. Businesses and employers: Employers and businesses that do not provide flexible working arrangements or promote alternative commuting options, such as telecommuting or flexible working hours, can contribute to peak-hour congestion. Lack of collaboration between businesses to stagger working hours or promote carpooling options can also exacerbate the problem.
5. Public transport operators: Public transport providers, such as bus companies, matatu operators, and commuter train services, are responsible for providing efficient and reliable services to encourage people to choose public transportation over private vehicles. Inadequate service quality, limited coverage, and lack of integration between different modes of public transport can contribute to congestion as people opt for private vehicles.

Addressing the responsibilities of these key stakeholders is crucial for mitigating traffic congestion in Nairobi. Collaborative efforts between government authorities, urban

planners, businesses, transportation agencies, and the public are necessary to implement effective transportation policies, infrastructure improvements, and behavior change campaigns to alleviate the problem.

Now that you have the preliminary information you need, you'll continue your web research to find some numbers, or quantifiable information, to help describe your problem:

What numerical data can you find that is relevant to your problem? Be sure to use your own words and also cite (give credit to) your sources.

Example 1:

According to Chat GPT, there are about 880 white rhinos currently living in Kenya. This population is very small, and they are critically endangered.

Example 2:

Johannesburg has approximately 5.8 million residents (per ChatGPT) and had approximately 4.7 million international overnight visitors in 2019 (according to the South African Tourism Annual Report for 2019/2020).

12. Approximately how many people (and/or animals) are **directly** impacted by this problem? Explain.

I onboarded Chat GPT as my research partner on this problem statement and according to chatGPT

1. Population of Nairobi: As of 2021, the estimated population of Nairobi was around 4.4 million people (source: World Population Review).

2. Vehicle ownership: Nairobi has a high vehicle ownership rate, with approximately 700,000 registered vehicles (source: Kenya National Bureau of Statistics). While not all vehicle owners may be directly impacted by traffic congestion, it gives us an indication of the number of vehicles that contribute to congestion.

3. Commuters: Nairobi serves as a major economic hub, attracting commuters from neighboring towns and suburbs. It is estimated that hundreds of thousands of people commute into the city for work daily, adding to the congestion during peak hours.

4. Public transportation users: Nairobi has a significant reliance on public transportation, including buses, matatus (minibusses), and commuter trains. The

number of people using public transportation and being affected by congestion is substantial.

Considering these factors, it is reasonable to estimate that hundreds of thousands, if not millions, of people are directly impacted by traffic congestion in Nairobi on a daily basis. The exact number may vary depending on the specific location, time of day, and other factors.

13. Approximately how many people (and/or animals) are **indirectly** impacted by this problem? Explain.

1. Businesses: Traffic congestion affects businesses in terms of delayed deliveries, increased transportation costs, and reduced productivity. Nairobi is a major economic center, and the number of businesses affected by congestion is significant. According to the Kenya National Bureau of Statistics, Nairobi County had approximately 411,000 registered businesses in 2019.

2. Supply chain and logistics: Traffic congestion can disrupt supply chains and logistics operations, leading to inefficiencies and increased costs. The indirect impact extends to suppliers, distributors, and retailers who rely on timely transportation of goods.

3. Environmental impact: Traffic congestion contributes to increased emissions, air pollution, and carbon footprint. This impact affects the overall air quality and environmental health of Nairobi, indirectly impacting the well-being of its residents and ecosystems.

4. Time loss and productivity: Indirectly, traffic congestion leads to time wasted by commuters, resulting in reduced productivity and decreased quality of life. Quantifying the exact number of people impacted indirectly in terms of time loss and reduced productivity is challenging.

Considering these factors, it is reasonable to estimate that hundreds of thousands, if not millions, of people are indirectly impacted by traffic congestion in Nairobi. The exact number may vary depending on the specific sectors and aspects considered. It's important to note that these estimates are based on available information and may not capture the entire scope of indirect impacts.

14. What other numerical data can you share that is relevant to your problem? What can you find out about its size and scope? What can be measured? (For example, the amount of trash produced in Nairobi each day, the number of people without access to clean water, etc.)

1. Commuter Population: Nairobi has a significant commuter population, with a large number of people traveling to and from the city for work and other activities. According to the Kenya National Bureau of Statistics, Nairobi had a population of approximately 4.4 million people in 2019. However, during weekdays, the population significantly increases as commuters from nearby areas travel to Nairobi for work.

2. Vehicle Ownership: The number of registered vehicles in Nairobi provides insights into the size and scope of the transportation system. According to the National Transport and Safety Authority (NTSA), as of 2020, Nairobi had over 1.8 million registered vehicles, including private cars, public transport vehicles, motorcycles, and trucks.

3. Travel Time: Measuring the average time spent in traffic can help understand the severity of congestion. A study conducted by the Nairobi Metropolitan Area Transport Authority (NAMATA) in 2017 found that commuters in Nairobi spent an average of 62 minutes in traffic each day, with some peak-hour trips taking significantly longer.

4. Road Length and Density: Assessing the road infrastructure in Nairobi provides insights into the physical size and complexity of the transportation network. Nairobi has an extensive road network, with major highways, arterials, and local roads. The Kenya Roads Board reported that Nairobi County had over 4,000 kilometers of classified roads in 2018.

5. Congestion Index: The TomTom Traffic Index ranks cities based on their congestion levels. In the 2020 index, Nairobi ranked among the top 100 most congested cities globally, indicating the significant scale of the problem.

6. Economic Cost: Quantifying the economic cost of traffic congestion provides a measure of its impact on productivity and efficiency. A study by the Nairobi Metropolitan Area Transport Authority estimated that Nairobi loses approximately 100 million Kenyan shillings (around \$930,000) per day due to traffic congestion.

By considering these numerical data points, we can gain a better understanding of the size, scope, and impact of the traffic congestion problem in Nairobi.

Step 4: Describe Your Solved State

Without having to come up with *how* to solve the problem, describe what the desired, solved state looks like. Please use numbers wherever possible, and make your solved state specific and measurable.

Example 1:

There would be a population of 10,000 healthy and protected white rhinos living in the wild in Kenya.

Example 2:

All 5.8 million residents of Johannesburg would have affordable and consistently available power from clean energy sources, 99.5% of the time.

15. If the problem were addressed/solved, what would reality be like?

In the desired, solved state where the traffic problem in Nairobi is addressed using current technology and computer vision, reality would be transformed in the following ways:

1. Reduced Travel Time: The average travel time for commuters would be significantly reduced. For example, the average daily commute time could be reduced from the current 62 minutes to around 30 minutes or less.
2. Improved Traffic Flow: The flow of traffic would be smoother, with reduced congestion and fewer traffic jams. This could be measured by a substantial decrease in the number of traffic bottlenecks and areas of gridlock across the city.
3. Increased Average Vehicle Speed: The average speed of vehicles on major roadways would improve. For instance, the average speed could increase from the current 15 km/h to around 30 km/h, allowing for faster and more efficient travel.

4. Enhanced Road Safety: With advanced computer vision technologies, there would be a significant improvement in road safety. The number of road accidents and incidents would decrease, resulting in fewer injuries and fatalities.

5. Optimal Traffic Signal Control: Computer vision algorithms could optimize traffic signal control, leading to better synchronization and coordination of traffic signals. This would result in smoother traffic flow, minimizing unnecessary stops and delays at intersections.

6. Efficient Traffic Management: Real-time traffic monitoring and analysis using computer vision would enable authorities to respond swiftly to incidents, accidents, and congestion hotspots. This would facilitate proactive traffic management, reducing the time taken to clear incidents and restore normal flow.

7. Increased Road Capacity Utilization: Advanced technologies could help maximize the utilization of existing road capacity. Through intelligent traffic management, the same road infrastructure could accommodate a higher volume of vehicles without significant congestion.

8. Improved Air Quality: By reducing traffic congestion and optimizing traffic flow, there would be a positive impact on air quality. The reduction in vehicle idle time and smoother traffic movement would contribute to lower emissions and improved environmental conditions.

By achieving these specific and measurable outcomes, the solved state would provide a more efficient, safer, and sustainable transportation system in Nairobi, significantly enhancing the quality of life for residents and promoting economic productivity.

16. Are there other benefits that would come from your problem being solved? Name at least one.

One additional benefit that would come from solving the traffic problem in Nairobi is the reduction in fuel consumption and carbon emissions. With improved traffic flow and reduced congestion, vehicles would spend less time idling or moving at slow speeds, resulting in lower fuel consumption. This, in turn, would lead to a decrease in carbon emissions and contribute to a greener and more environmentally sustainable

city. By addressing the traffic problem, Nairobi would take a step towards mitigating its environmental impact and promoting a more sustainable transportation system.

Step 5: Clarify Your Problem Scope

You are more effective at solving a problem when you know where its limits are. That is, when you know what is “in scope” and “out of scope.” For this reason, it is important to list what is out of scope, or NOT included as part of your problem definition.

Example 1:

The scope of the problem does not cover any other animal species besides white rhinos. It does not include white rhinos outside of Kenya.

Example 2:

The scope of the problem does not include any businesses or people outside of the legally-defined Johannesburg city limits. It does not apply to tourists or visitors staying for less than 1 year in Johannesburg.

17. What is NOT in scope for your problem?

In the scope of solving the traffic jams in Nairobi, the following aspects are considered out of scope:

1. Transportation infrastructure development: While the problem acknowledges the impact of inadequate infrastructure on traffic congestion, the scope does not include planning, designing, or implementing new transportation infrastructure projects.
2. Policy and governance changes: Although political factors were identified as contributing to the problem, the scope does not involve proposing or implementing changes in government policies or regulations related to transportation or urban planning.
3. Urban planning and zoning: While urban sprawl and development were identified as factors contributing to traffic congestion, the scope does not include comprehensive urban planning, zoning regulations, or land-use decisions.
4. Behavioral changes: Although cultural and social norms were discussed, the scope does not cover initiatives to change individual behavior or travel patterns, such as

promoting public transportation usage or carpooling.

5. Economic policies: While economic factors were considered, the scope does not encompass proposing or implementing economic policies, such as congestion pricing or road user charges.

The focus of the problem scope is on leveraging technology, particularly computer vision and AI, to address traffic congestion in Nairobi. It seeks to explore solutions within the realm of technological advancements and data-driven approaches to optimize traffic flow and alleviate congestion.

Step 6: Areas for Learning

What do you not know or understand that you would like to know more about? This can be anything related directly or indirectly to your problem. Let your curiosity run wild!

Example 1:

I'd like to know where most of the demand for rhino horn is coming from. I'd like to know who are the primary buyers and who is behind the trafficking of rhino horn. I'd like to know how long the average rhino's lifespan is. I'd like to know how many babies a typical female rhino has, and how many babies typically survive into adulthood. I'd like to know more about what diseases impact rhinos. I'd like to know more about the kinds of habitats that rhinos thrive in. I'd like to understand what international organizations do the best job supporting wildlife conservation and what their practices are. I'd like to know what models of community involvement have been most successful in keeping wildlife safe and thriving. I'd like to understand how much land is available in Kenya for rhinos to roam.

Example 2:

I'd like to better understand the utility company Eskom and its history. I'd like to understand why Eskom has failed to plan properly to update its infrastructure. I'd like to understand the relationship between Eskom and the South African government. I'd like to know if there are private utility companies providing competition to Eskom. I'd like to know what the latest breakthroughs are in solar power. I'd like to know what other possible energy sources might be made available in Johannesburg. I'd like to know how much energy tourists and temporary visitors use. I'd like to better understand the process of how limited energy supply gets allocated to people and businesses. I'd like to better understand the economic impact to people and businesses of not having power.

18. What else would you like to know or understand better? (It can be anything related to your problem.) List 5-10 things.

I am curious to learn more about the implementation of high-speed trains in Nairobi and their potential impact on traffic congestion. Specifically, I would like to understand the following aspects:

1. Reduction in travel time: I am interested in quantifying the time savings that high-speed trains can offer compared to commuting by road. It would be insightful to compare the average travel time for a specific route between a high-speed train and private vehicles during peak hours.
2. Passenger capacity: I would like to explore the passenger capacity of high-speed trains and compare it to the number of vehicles that can be accommodated on expanded highways. This information would provide insights into the potential reduction in the number of vehicles on the roads and its impact on traffic congestion.
3. Ridership projections: It would be valuable to study the ridership projections for high-speed trains in Nairobi and estimate the potential number of commuters who would shift from private vehicles to train transportation. Understanding the potential shift in mode of transport can help quantify the impact on traffic volume.
4. Environmental benefits: I am curious to learn about the environmental benefits of high-speed trains compared to increased highway expansion. Quantifying the reduction in greenhouse gas emissions, air pollution, and fuel consumption resulting from a modal shift to high-speed trains would provide valuable insights.
5. Integration with existing transportation systems: I would like to know more about the integration of high-speed trains with existing transportation infrastructure in Nairobi, such as bus networks and metro systems. Understanding how these modes of transport can complement each other and provide seamless connectivity is essential.

By delving into these areas of interest, we can gain a deeper understanding of the potential benefits and feasibility of implementing high-speed trains in Nairobi and how they can contribute to alleviating traffic congestion while providing a sustainable transportation solution.

Step 7: Problem Statement

This step is the culmination of all you have done in Part A. You will synthesize the work you have done above to create a problem statement of 150 - 250 words. This should be in narrative form, 2-4 paragraphs, and should NOT use bullet points.

Your problem statement should:

- Provide a succinct description of the problem **in the first sentence**.
- Indicate specific population affected
- Explain the impact (cost, time, environmental, personal) and why the problem matters.
- Explain what reality would look like if the problem were solved. The gap that exists between present reality and the desired outcome should be clear.

Please cite (give credit to) where your information came from directly in your statement. Avoid word-for-word quoting and instead paraphrase (use your own words), as modeled in the example. Also list your sources and their urls (web addresses) at the end.

Example :

Kenya's white rhinos are in critical danger of extinction. There are currently about 880 white rhinos in the country of Kenya, per Wikipedia. According to Chat GPT, Rhinos are considered a keystone species, meaning they have a disproportionately large impact on their ecosystem compared to their population size. Rhinos help shape their environment by influencing vegetation growth and acting as seed dispersers, which creates habitat for other species (per ChatGPT).

The extinction of white rhinos would have cascading effects on other plant and animal species in their habitat. According to the Kenya Wildlife Service, rhinos' presence in reserves and parks bring millions of tourists each year, contributing to local economies and supporting conservation efforts. Once a species goes extinct, it is gone forever. The extinction of rhinos would represent the loss of millions of years of evolutionary history, and unique genetic diversity that science has yet to fully understand and benefit from (per ChatGPT).

My problem would be considered solved when the population of wild, white rhinos in Kenya reaches 10,000, and when all imminent threats to their population including poaching and habitat destruction are not present. If this were the reality, it would create ecosystem balance, create large revenues from ecotourism, preserve important cultural symbols, and allow for genetic diversity that could benefit humanity in ways we may not yet fully understand.

Sources:

Kenya Wildlife Services Annual Report 2017, <https://www.kws.go.ke/content/annual-reports>

ChatGPT, <https://chat.openai.com/>

"White Rhinoceros", Wikipedia, https://en.wikipedia.org/wiki/White_rhinoceros

19. My problem is statement is:

The traffic congestion problem in Nairobi, Kenya has reached critical levels, impacting the daily lives of millions of residents and businesses. Commuters in Nairobi experience significant delays, frustration, and lost productivity due to excessive time spent in traffic. The lack of efficient transportation infrastructure, inadequate alternative modes of transport, and urban sprawl have contributed to this issue, resulting in a burden on the city's economy, environment, and overall quality of life. According to data from the Nairobi Metropolitan Area Transport Authority, traffic congestion costs the city an estimated 50 billion Kenyan shillings (approximately \$460 million) annually in lost productivity and fuel consumption [1]. This problem is of paramount importance as it affects the well-being, productivity, and mental health of individuals, contributes to increased greenhouse gas emissions and air pollution, and hampers the city's economic growth.

If the traffic congestion problem in Nairobi were effectively addressed and high-speed trains were implemented as a sustainable transportation solution, the city would experience a transformative change. With reduced travel times and increased passenger capacity, high-speed trains would alleviate the heavy reliance on private vehicles and provide a more efficient and reliable mode of transportation. The implementation of high-speed trains would not only significantly reduce travel time for commuters but also enhance connectivity, accessibility, and safety. A well-integrated and comprehensive transportation system, including high-speed trains, would lead to a substantial reduction in traffic congestion, promoting economic growth, improving air quality, and enhancing the overall quality of life for residents.

Sources:

[1] Nairobi Metropolitan Area Transport Authority. "Economic Impact of Traffic Congestion." <http://www.namata.go.ke/>

[2] World Bank. "Kenya National Urban Transport Improvement Project."

<https://documents1.worldbank.org/curated/en/702181563299068935/pdf/Kenya-National-Urban-Transport-Improvement-Project.pdf>

20. Please list all sources that you used to form your problem statement:

- [1] Nairobi Metropolitan Area Transport Authority. "Economic Impact of Traffic Congestion."
"Nairobi Integrated Urban Development Master Plan."
<http://www.namata.go.ke/>
- [2] World Bank. "Kenya National Urban Transport Improvement Project."
<https://documents1.worldbank.org/curated/en/702181563299068935/pdf/Kenya-National-Urban-Transport-Improvement-Project.pdf>
- [3] KNBS. "Kenya National Bureau of Statistics"
<https://www.knbs.or.ke/>
- [4] The TomTom Traffic Index. "Congestion Ranking." Accessed from:
https://www.tomtom.com/en_gb/traffic-index/
- [5] National Transport and Safety Authority (NTSA)
<https://www.nts.go.ke/>
- [6] Kenya Roads Board. "Road Infrastructure."
<http://www.krb.go.ke/>
- [7] ChatGPT
<https://chat.openai.com>



Please go back to Canvas and continue with your learning content. You will be prompted on when to return to complete Section B.

SECTION B: Research Questions & Hypothesis

IMPORTANT: Complete this section AFTER completing the Canvas Modules *Asking Effective Questions* and *Web Research*.

Step 8: Research Questions

Based on what you have learned so far and on 'Step 6: Areas for Learning' from this worksheet, come up with 3 research questions. **Research questions should be complex enough that they**

can't be answered by a single Google search. If appropriate, form a hypothesis that your research may confirm or reject. (As a reminder, a hypothesis is a prediction of how you think your research will answer your research question. It is your best guess. If you truly have absolutely no idea, state "not applicable.")

Example Research Question #1:

What are some ways can we increase rhino populations?

Hypothesis:

Rhino populations will be increased by creating more open spaces for them to roam, increasing their protection, increasing international interest in them, and other reasons I have yet to uncover.

Example Research Question #2:

Which organizations have been effective at wildlife conservation and what practices do they use?

Hypothesis:

Not applicable; I don't know.

Example Research Question #3:

How many babies can a typical female white rhino have in her lifetime, and what are the reasons a female may not have high fertility?

Hypothesis:

A typical female white rhino can have 5 babies in her lifetime, and fertility may be affected by diet, amount of grazing territory, poaching, stress, mate availability, and other reasons I have yet to uncover.

21. Research question #1:

How much time can high-speed trains save for commuters compared to private vehicles during peak hours in Nairobi?

Hypothesis (if applicable):

Hypothesis: The implementation of high-speed trains in Nairobi will significantly reduce travel time, resulting in a substantial time savings for commuters compared to private vehicles during peak hours.

22. Research question #2:

What is the passenger capacity of high-speed trains, and how does it compare to the number of vehicles that can be accommodated on expanded highways in Nairobi?

Hypothesis (if applicable):

Hypothesis: High-speed trains in Nairobi have a higher passenger capacity compared to the number of vehicles that can be accommodated on expanded highways. This suggests that the introduction of high-speed trains can lead to a significant reduction in the number of vehicles on the roads and contribute to alleviating traffic congestion.

23. Research question #3:

What are the ridership projections for high-speed trains in Nairobi, and how many commuters are likely to shift from private vehicles to train transportation?

Hypothesis (if applicable):

Hypothesis: The availability of high-speed trains in Nairobi will attract a considerable number of commuters, resulting in a notable modal shift from private vehicles to train transportation. This shift is expected to lead to a decrease in traffic volume and contribute to reducing traffic congestion.

Additional areas for understanding:

4. Cost-effectiveness: The implementation and operation of high-speed trains will be more cost-effective compared to expanding highways in Nairobi.

5. Safety and reliability: High-speed trains will have a superior safety record and reliability measures compared to road transportation.

6. Integration challenges: Integrating high-speed trains with existing transportation systems in Nairobi, such as bus networks and metro systems, may present potential challenges and considerations.

7. Social impact: The introduction of high-speed trains will have a positive impact on the daily lives and commuting experiences of individuals in Nairobi.

8. Economic impact: Implementing high-speed trains will bring economic benefits, such as job creation, economic growth, and real estate development around train stations. However, there may also be some drawbacks to consider.

9. Accessibility and equity: High-speed trains will improve accessibility for different socioeconomic groups in Nairobi, with measures in place to ensure equitable access to this mode of transport.

10. Lessons from other cities: Examining the implementation of high-speed trains in other cities worldwide will provide insights into their impact on traffic congestion, urban development, and sustainable transportation.

Sources:

1. Kenya Railways Corporation - <https://www.krc.co.ke/>
2. Nairobi Metropolitan Area Transport Authority (NAMATA) - <http://namata.go.ke/>
3. Kenya National Bureau of Statistics - <https://www.knbs.or.ke/>
4. National Transport and Safety Authority (NTSA) - <https://www.nts.go.ke/>
5. International Union of Railways - <https://www.uic.org/>
6. United Nations Environment Programme (UNEP) - <https://www.unep.org/>



Please go back to Canvas and continue with your learning content. You will be prompted on when to return to complete Section C.

PART C: Mini-Research Report

IMPORTANT: Complete this section AFTER completing the Canvas Module *Web Research Executive Summary*.

In 200-300 words, provide an executive summary of your research. You should be synthesizing information from multiple sources. Provide answers and explanations for the 3 questions you investigated and your key research findings. This should be in a narrative format (no bullet points), and be at least 3 paragraphs long.

Please use at least 3 different online sources such as ChatGPT, organizational websites, wikipedia, etc.. Please cite (give credit to) where your information came from directly in your statement. Avoid word-for-word quoting and instead paraphrase (use your own words), as modeled in the example. Also list your sources and their urls (web addresses) at the end.

Example Executive Summary:

According to ChatGPT, increasing rhino populations requires multiple approaches. These approaches include: anti-poaching measures; habitat protection and management; translocation and reintroduction; captive breeding and reintroduction; international cooperation between governments, conservation organizations, and communities; community engagement and education; and demand reduction.

According to ChatGPT, the following organizations are well known for wildlife conservation and protection: World Wildlife Fund (WWF), International Union for Conservation of Nature (IUCN), Wildlife Conservation Society (WCS), African Wildlife Foundation (AWF), Pathera, Conservation International, and The Jane Goodall Institute. These organizations engage in different practices to help wildlife. As one example, World Wildlife Fund, according to their webpage, engages in protecting land where animals roam, educating communities on the importance of protecting wildlife, and shutting down markets in Asia where ivory is sold.

Another example is African Wildlife Foundation. According to their website, they focus their efforts conservation education, anti-poaching efforts, habitat restoration, and policy advocacy. The Jane Goodall Institute is a final example. According to their website, they focus on environmental education, and promoting sustainable livelihoods through conservation efforts. There appears to be several ways that reputable wildlife conservation organizations go about accomplishing their goal.

According to the International Rhino Foundation, a typical rhino pregnancy lasts around 16 months. According to the Maryland Zoo, Female white rhinos are able to start producing offspring when they are about 6 years old. A female rhino can give birth to one calf about once every 2-3 years. According to ChatGPT, a female rhino could give birth to as many as 14 offspring in her lifetime, but this is uncommon. Usually, conditions do not support this.

Sources:

*African Wildlife Foundation, <https://www.awf.org/strategic-vision-2020-2030>
ChatGPT, <https://chat.openai.com/>*

Jane Goodall Institute, <https://janegoodall.org/our-impact/>

International Rhino Foundation <https://rhinos.org/blog/rhino-moms/>

Maryland Zoo, <https://www.marylandzoo.org/animal/southern-white-rhinoceros/>

World Wildlife Fund, <https://www.worldwildlife.org/initiatives/wildlife-conservation>

24. Executive Summary:

In the face of growing traffic congestion challenges in Nairobi, there is a need for innovative solutions to effectively manage and alleviate the problem. This research focuses on exploring the potential of high-speed trains as a sustainable transportation solution and the integration of computer vision and AI technologies to optimize traffic management.

The first research question examines the time-saving benefits of high-speed trains compared to private vehicles during peak hours in Nairobi. By analyzing travel time data, it is hypothesized that high-speed trains will significantly reduce commute times, providing substantial time savings for commuters.

The second research question investigates the passenger capacity of high-speed trains and its comparison to expanded highways in Nairobi. The hypothesis suggests that high-speed trains have a higher passenger capacity than expanded highways, indicating the potential to reduce the number of vehicles on the roads and alleviate traffic congestion.

The third research question focuses on the ridership projections for high-speed trains in Nairobi and the potential modal shift from private vehicles to train transportation. It is hypothesized that high-speed trains will attract a considerable number of commuters, resulting in a notable decrease in traffic volume.

Furthermore, this research explores additional areas related to the implementation of high-speed trains and the integration of computer vision and AI technologies. These areas include cost-effectiveness, safety and reliability, integration challenges with existing transportation systems, social impact, economic benefits, accessibility and equity, and lessons from other cities.

To achieve these research objectives, a combination of qualitative and quantitative methods will be employed, including data analysis, surveys, and interviews. Multiple sources will be utilized, including the Kenya Railways Corporation, Nairobi Metropolitan Area Transport Authority (NAMATA), Kenya National Bureau of Statistics, National Transport and Safety Authority (NTSA), International Union of Railways, and United Nations Environment Programme (UNEP).

By addressing these research questions and considering the potential of computer

vision and AI technologies, this research aims to provide valuable insights for policymakers, urban planners, and transportation authorities in Nairobi, facilitating informed decision-making to tackle traffic congestion and pave the way for a more efficient and sustainable transportation system.

25. Please list all sources that you used to form your executive summary:

Sources:

1. Kenya Railways Corporation - <https://www.krc.co.ke/>
2. International Union of Railways - <https://www.uic.org/>
3. Nairobi Metropolitan Area Transport Authority (NAMATA) - <http://namata.go.ke/>
4. Kenya National Bureau of Statistics - <https://www.knbs.or.ke/>
5. National Transport and Safety Authority (NTSA) - <https://www.nts.go.ke/>
6. International Union of Railways - <https://www.uic.org/>
7. United Nations Environment Programme (UNEP) - <https://www.unep.org/>



Please go back to Canvas and continue with your learning content. Return to Part D after you have completed your City Hub Activities.

PART D: Hub Activities Report

IMPORTANT: Complete this section AFTER completing the Canvas Lessons “Hub Activity: Problem Statement” and “Hub Activity: Web Research.”

Hub Activity #1: Problem Statement

Please report on your process of getting peer feedback.

26. Who reviewed your problem statement (item #19)? (Give the first and last names of your 2 peers.)

Ann Julie Njeri

Simon Derrick Njenga

Willy's Wanyiri

27. In brief, what feedback did they give to you?

That my problem statement would solve a problem plaguing the city of Nairobi

28. Was their feedback useful to you? Did it feel kind? Why or why not?

Yes, their feedback was insightful and they also provided more sources of where I could gain insight in developing my problem statement.

29. Who did you give feedback to? (Give the first and last names of 2 peers- they may be the same or different peers from above.)

Ann Julie Njeri

Willy's Wanyiri

30. Do you feel that you gave useful and kind feedback to your peers? Why or why not?

Yes, we analyzed the problem statements and provided in depth feedback and also reflected on the problems they'd solve

Hub Activity #2: Web Research

Please report on your process of getting peer feedback.

31. Who reviewed your executive summary (item #24)? (Give the first and last names of your 2 peers.)

Willy's Wanyiri

Ann Julie Njeri

32. In brief, what feedback did they give to you?

Constructive feedback in that providing a solution to traffic congestion would improve the city's outlook as an east african Tech hub.

33. Was their feedback useful to you? Did it feel kind? Why or why not?

Yes it was useful feedback and also kind. They added that it's up to the youth of Kenya to develop solutions to the daily challenges that we face.

34. Who did you give feedback to? (Give the first and last names of 2 peers- they may be the same or different peers from above.)

Ann Julie Njeri

Willy's Wanyiri

35. Do you feel that you gave useful and kind feedback to your peers? Why or why not?

Yes, for one of the Executive summary, it looks to address the gap in private vs public universities in Kenya. Such a grand challenge needs to be one of the key focus of the government policies.



After you updated earlier parts of the worksheet based on your feedback, please go back to Canvas and continue with your learning content. Return to Part E prior to submitting your milestone.

PART E: Daily 3 Challenge Report

Please answer the following questions honestly. There are no wrong answers! This is your opportunity for self-reflection.

36. How many days out of the past 7 did you do 20 minutes of movement?

7 days

37. How many days out of the past 7 did you write 3 morning pages?

4 days

38. Overall, how do you feel you are doing on building your Daily 3 habits? Explain.

I am off to a great start at building my daily habits. There's room for improvement and I know that over the next few months I'll incorporate this into my routine and grow.

39. What is your biggest barrier to staying motivated?

My busy schedule at times gets in the way and also procrastination.

40. If you have been practicing at least some movement and/or morning pages:

YES

A) What effect (if any) have you noticed on your mood, focus, and productivity?

Working out ensures that I have energy throughout my day, improved my focus and boosted my creativity. I'm able to get through my tasks faster now.

B) What helps you stay motivated?

The feeling of accomplishing my goal in time gives me great joy.

41. If you haven't been practicing any movement and morning pages, what would motivate you to get started?

Finding a peer to keep me motivated.

Once you have completed this worksheet, export/convert to .pdf, rename it per the instructions, and upload to Canvas as your Milestone 3 Submission. Celebrate a job well done!